POSITIVE BLACK HOLE DEGENERACIES FOR DYONS OF HIGHER TORSION

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Abstract. Sen's argument says that a single-centred quarter-BPS black hole carries a positive index. For dyons of torsion $r > 1$ in string theory on $K3 \times T^2$ we prove this for every r when the magnetic charge has $P^2/2 = 1$ or 2. For every magnetic charge, odd torsion follows from the primitive case and even torsion from one inequality between primitive indices.

Keywords: black hole degeneracies; non-primitive dyons; mock Jacobi forms.

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1. Introduction

A quarter-BPS dyon of $\mathcal{N} = 4$ string theory in four dimensions carries an electric charge vector Q and a magnetic charge vector P in the Narain lattice $\Gamma^{22,6}$, and its protected index is a sixth helicity supertrace. When the charges support a single-centred black hole, Sen observed that the only states not belonging to the horizon are the zero modes of the broken supersymmetries, and that the spherical symmetry of the horizon then forces the horizon states to be bosonic [21]. The index of a single-centred black hole, normalised so as to strip off the fermion zero modes, must therefore be positive.[1] On the microscopic side the index is a Fourier coefficient of the inverse of a Siegel modular form, and the conjecture becomes a statement about the signs of these coefficients.

For charges of torsion one, that is with $\gcd(Q \wedge P) = 1$, the partition function is the inverse of the Igusa cusp form Φ_{10} [13, 22]. Dabholkar, Murthy and Zagier showed that, once the contribution of two-centred bound states is removed, the generating function of the single-centred indices of magnetic charge $m = P^2/2$ is a mock Jacobi form ψ_m^F; Bringmann and Murthy used explicit formulas for ψ_1^F and ψ_2^F to prove the positivity conjecture for $m = 1$ and $m = 2$ and every electric charge [4].[2]

Most charges are not primitive. A dyon has torsion $r = \gcd(Q \wedge P)$, the greatest common divisor of the 2×2 minors $Q_i P_j - Q_j P_i$, and for $r > 1$ the partition function is no longer $1/\Phi_{10}$. Banerjee, Sen and Srivastava proposed the partition function of the family $\mathcal{B}_r$ of dyons with Q equal to r times a primitive vector [6, 7], and Dabholkar, Gomes and Murthy derived the same formula from the microscopic theory [11]. The index of a dyon of torsion r is a weighted sum of indices of torsion one with the weights s running over the divisors of r, and with signs. The signs are the whole difficulty: for even r some of the terms are negative, so positivity of the primitive indices does not by itself give positivity at torsion r. As far as we know the sign of the single-centred index at torsion $r > 1$ has not been settled for any family of charges.

[1] The argument and the normalisation are explained in Sections 1 and 2 of [21]. The relation between the index and the entropy of the black hole goes back to the entropy function formalism reviewed in [20], and the relation between index and degeneracy for supersymmetric black holes is derived in [12]. In frames where the classical solution carries hair outside the horizon one must first remove the hair; for type II string theory on $K3 \times T^2$ in the heterotic frame no further hair is present, see [9].

[2] A review of these mock Jacobi forms and their role in black hole counting is [17]. The construction of Φ_{10} as an automorphic correction of a Lorentzian Kac–Moody algebra is in [15].

Write $n = Q^2/2$, $\ell = Q \cdot P$, $m = P^2/2$ and $\Delta = 4mn - \ell^2 = Q^2P^2 - (Q \cdot P)^2$. Let $d^*_r(Q, P)$ be the index of a dyon $(Q, P) \in \mathcal{B}_r$ computed with Sen's attractor contour (Definition 2.1). Our main result is the following.

Theorem A. *Let $r \geq 1$ and let $(Q, P) \in \mathcal{B}_r$ with $P^2/2 \in \{1, 2\}$ and $Q^2P^2 - (Q \cdot P)^2 > 0$. Then $d^*_r(Q, P) > 0$.*

The proof has two parts. The first is a reduction that holds for every magnetic charge.

Theorem B. *Fix $m \geq 1$ and assume that the single-centred index of torsion one satisfies $d^*(n', \ell', m) > 0$ whenever $4mn' - \ell'^2 > 0$. Then $d^*_r(Q, P) > 0$ for every odd r and every $(Q, P) \in \mathcal{B}_r$ with $P^2/2 = m$ and $\Delta > 0$. If in addition*

$$d^*(4n', 2\ell', m) > 2\, d^*(n', \ell', m) \qquad \text{for all } \ell' \text{ odd with } 4mn' - \ell'^2 > 0, \tag{1}$$

then $d^*_r(Q, P) > 0$ for every $r \geq 1$.

The second part proves the doubling inequality (1) for $m = 1$ and $m = 2$.

Theorem C. *The doubling inequality (1) holds for $m = 1$ and for $m = 2$.*

Theorem A follows from Theorems B and C together with the positivity of the primitive indices for $m = 1, 2$ proved in [4].

The idea behind Theorem B is a pairing. In the torsion sum the term of a divisor s is negative only when s is even and ℓ/s is odd, and then the term of $s/2$ is positive and involves the charge $(4n/s^2, 2\ell/s, m)$, whose discriminant is four times that of the negative term. Since the single-centred index grows like $\exp(\pi\sqrt{\Delta})$, quadrupling the discriminant doubles the exponent, and one expects (1) to hold with an enormous margin. Figure 4 shows that the ratio of the two sides of (1) is at least 325 in every case we computed, and grows quickly with the discriminant.

The proof of Theorem C is elementary. We write the two sides of (1) through the q-series of Bringmann and Murthy, drop the class number terms in the direction that weakens the inequality, and compare the remaining coefficients through two facts about coloured partitions: the bound $p_k(N + 1) \leq (k + 1)\, p_k(N)$ for the number of k-coloured partitions, and the factorisation $(q; q)_\infty^{-k} = \Psi(q)^k\, (q^4; q^4)_\infty^{-k}$, where Ψ counts partitions into parts not divisible by four. The factorisation matches the coefficient of q^{4N} on the left of (1) with the coefficient of q^N on the right term by term, and the remaining estimates are explicit inequalities between binomial coefficients and divisor sums. A finite table of exact values covers the smallest discriminants: $D' \leq 31$ for $m = 1$ and $D' \leq 39$ for $m = 2$, where D' is the discriminant of the negative term.

The paper is organised as follows. Section 2 recalls the partition functions and derives the torsion formula for the single-centred index with its signs. Section 3 proves Theorem B. Section 4 contains the two partition lemmas. Sections 5 and 6 prove the doubling inequality for $m = 1$ and $m = 2$, and Section 7 completes the proof of Theorem A and discusses general m. Appendix A derives the q-series from the formulas of [4], Appendix B proves the coefficient bounds, and Appendix C lists the exact values used.

Notation. We write $q = e^{2\pi i\tau}$, $\mathbf{e}(x) = e^{2\pi ix}$ and $(q; q)_\infty = \prod_{j\geq 1}(1 - q^j)$. For a power series $F = \sum a(j)q^j$ we write $\big[F\big]_j = a(j)$, with $\big[F\big]_j = 0$ for $j < 0$, and $F \succeq 0$ when all coefficients are non-negative. For $k \geq 1$ let

$$P_k(q) = (q; q)_\infty^{-k} = \sum_{N\geq 0} p_k(N)\, q^N,$$

so that $p_k(N)$ is the number of k-coloured partitions of N. The Eisenstein series are $E_2 = 1 - 24\sum \sigma_1(j)q^j$, $E_4 = 1 + 240\sum \sigma_3(j)q^j$ and $E_6 = 1 - 504\sum \sigma_5(j)q^j$, the operator $D = q\,\frac{d}{dq}$

acts on q-series, and $H(N)$ is the Hurwitz class number, with $H(0) = -1/12$ and $H(N) > 0$ for $N \geq 1$ [16, 23]. We use the theta series

$$\theta_0 = \sum_{k\in\mathbb{Z}} q^{k^2}, \qquad T = \sum_{k\geq 0} q^{k(k+1)}, \qquad V = \sum_{k\geq 0} q^{k(k+1)/2}.$$

2. Charges, torsion and the single-centred index

2.1. **Torsion one.** Type IIB string theory on $K3 \times T^2$ is dual to the heterotic string on T^6, and in the heterotic frame the charges (Q, P) of a dyon are a pair of vectors in the even unimodular lattice $\Gamma^{22,6}$.[3] The T-duality invariants are $n = Q^2/2$, $\ell = Q \cdot P$ and $m = P^2/2$, and for torsion one they determine the charges up to T-duality. The partition function is

$$\frac{1}{\Phi_{10}(\tau, z, \tau')} = \sum_{m\geq -1} \psi_m(\tau, z)\, \mathbf{e}(m\tau'), \tag{2}$$

where Φ_{10} is the Igusa cusp form of weight 10 and ψ_m is a meromorphic Jacobi form of weight -10 and index m [4]. The variables τ, z, τ' are conjugate to n, ℓ, m.

Because Φ_{10} vanishes on the divisor $z = 0$ and its images, the Fourier coefficients of $1/\Phi_{10}$ depend on the imaginary parts of (τ, z, τ'). Following Sen [19, 21] and the contour prescription of Banerjee, Sen and Srivastava [8], the single-centred index is the coefficient taken on the contour fixed by the attractor values of the moduli.[4]

Definition 2.1. For (n, ℓ, m) with $\Delta = 4mn - \ell^2 > 0$ let $\mathcal{C}^*(n, \ell, m)$ be the contour

$$0 \leq \operatorname{Re}\tau, \operatorname{Re} z, \operatorname{Re}\tau' \leq 1, \qquad \operatorname{Im}\tau = 2m\Lambda, \quad \operatorname{Im}\tau' = 2n\Lambda, \quad \operatorname{Im} z = -\ell\Lambda,$$

with Λ large, and put

$$d^*(n, \ell, m) = (-1)^{\ell+1} \int_{\mathcal{C}^*(n,\ell,m)} \frac{\mathbf{e}(-n\tau - \ell z - m\tau')}{\Phi_{10}(\tau, z, \tau')}\, d\tau\, dz\, d\tau'. \tag{3}$$

For a dyon (Q, P) of torsion one, $d^*(Q^2/2, Q \cdot P, P^2/2)$ is its single-centred index.

The sign $(-1)^{\ell+1}$ accounts for the fermion zero modes and for the convention that counts bosonic minus fermionic supermultiplets [4, 21]. Two properties of d^* will be used. First, d^* is invariant under the S-duality group $SL(2, \mathbb{Z})$, which acts on (n, ℓ, m) as on the binary quadratic form $nx^2 + \ell xy + my^2$: the attractor contour moves covariantly with the charges [8, 19]. In particular

$$d^*(n, \ell, m) = d^*(n + k\ell + k^2 m,\ \ell + 2km,\ m) \quad (k \in \mathbb{Z}), \qquad d^*(n, \ell, m) = d^*(m, -\ell, n). \tag{4}$$

Second, d^* is a Fourier coefficient of a mock Jacobi form. Write $\psi_m = \psi_m^P + \psi_m^F$, where the polar part ψ_m^P is the Appell–Lerch sum that carries the poles of ψ_m, and the finite part

$$\psi_m^F(\tau, z) = \sum_{n,\ell} c_m^F(n, \ell)\, q^n \zeta^\ell, \qquad \zeta = \mathbf{e}(z),$$

is a mock Jacobi form of index m. Dabholkar, Murthy and Zagier proved that the single-centred index is read off from ψ_m^F [4, Section 3.1]: for a reduced triple, that is one with $|\ell| \leq m \leq n$,

$$d^*(n, \ell, m) = (-1)^{\ell+1} c_m^F(n, \ell). \tag{5}$$

[3] For the heterotic string and its relation to M-theory see [18]; for the geometry of $K3$ surfaces, their involutions and the $K3$-fibred Calabi–Yau threefolds behind the dual type II descriptions see [5, 2].

[4] At the attractor point of the charges no two-centred bound state exists, so only the single-centred black hole contributes; see [19] and [10] for the walls of marginal stability and the discrete attractor flow.

The coefficients of ψ_m^F satisfy the elliptic invariance of a Jacobi form of index m, so $c_m^F(n,\ell)$ depends only on $4mn-\ell^2$ and on ℓ modulo $2m$, and is even in ℓ.

Lemma 2.2. *Let $m \in \{1,2\}$. Then (5) holds for every (n,ℓ) with $4mn-\ell^2>0$.*

Proof. Both sides of (5) are unchanged under $(n,\ell)\mapsto(n+k\ell+k^2m,\ell+2km)$, the left side by (4) and the right side by elliptic invariance; note that $(-1)^\ell$ is unchanged as well. Choosing k we may assume $|\ell|\le m$. Since $4mn>\ell^2\ge 0$ we have $n\ge 1$. If $n\ge m$ the triple is reduced and (5) is the theorem quoted above. This settles $m=1$, and for $m=2$ leaves $n=1$ and $\ell\in\{0,\pm1,\pm2\}$. By (4), $d^*(1,\ell,2)=d^*(2,-\ell,1)$, and the right side equals $(-1)^{\ell+1}c_1^F(2,\ell)$ by the case $m=1$. It remains to see that $c_2^F(1,\ell)=c_1^F(2,\ell)$ for $\ell=0,1,2$. Both sides are computed from the closed formulas (12) and (16) below; the values of $(-1)^{\ell+1}c^F$ are

$$1127472, \qquad 561576, \qquad 50064$$

for $\ell=0,1,2$ on either side (Appendix C, Table 1).[5] □

2.2. **Torsion r.** Let $r\ge 1$. Following [6] let $\mathcal{B}_r$ be the set of dyons (Q,P) with Q equal to r times a primitive vector, P primitive, and Q/r and P spanning a primitive sublattice of $\Gamma^{22,6}$. Such dyons have torsion r, two of them with the same invariants are related by T-duality, and the invariants satisfy

$$\frac{Q^2}{2}\in r^2\mathbb{Z}, \qquad \frac{P^2}{2}\in\mathbb{Z}, \qquad Q\cdot P\in r\mathbb{Z}. \tag{6}$$

Banerjee, Sen and Srivastava [6, eqs. (3)–(7)] proposed, and Dabholkar, Gomes and Murthy derived from the microscopic conformal field theory [11, eqs. (3.24)–(3.28)], the following formula for the index $d(Q,P)$ of a dyon in $\mathcal{B}_r$:

$$d(Q,P)=(-1)^{Q\cdot P+1}\sum_{s|r} s\cdot s^3\int \frac{\mathbf{e}\big(-\frac12(\check\sigma Q^2+\check\rho P^2+2\check v\,Q\cdot P)\big)}{\Phi_{10}(\check\rho,s^2\check\sigma,s\check v)}\,d\check\rho\,d\check\sigma\,d\check v, \tag{7}$$

where the real parts run over $|\operatorname{Re}\check\rho|\le\frac12$, $|\operatorname{Re}\check\sigma|\le\frac{1}{2s^2}$, $|\operatorname{Re}\check v|\le\frac{1}{2s}$, and the imaginary parts are fixed in a region where the expansion converges.[6] The single-centred index $d_r^*(Q,P)$ is (7) with the imaginary parts placed on the attractor contour of (Q,P), which in these variables reads

$$\operatorname{Im}\check\rho=2n\Lambda, \qquad \operatorname{Im}\check\sigma=2m\Lambda, \qquad \operatorname{Im}\check v=-\ell\Lambda. \tag{8}$$

Proposition 2.3. *Let $(Q,P)\in\mathcal{B}_r$ with invariants (n,ℓ,m) and $\Delta>0$. Then*

$$d_r^*(Q,P)=\sum_{s|r} s\,(-1)^{\ell-\ell/s}\,d^*\Big(\frac{n}{s^2},\frac{\ell}{s},m\Big). \tag{9}$$

Every triple on the right has positive discriminant Δ/s^2.

Proof. By (6), n/s^2 and ℓ/s are integers for $s\mid r$. In the term of s substitute $\sigma=s^2\check\sigma$, $v=s\check v$ and $\rho=\check\rho$. The Jacobian is $d\check\sigma\,d\check v=s^{-3}d\sigma\,dv$, which cancels s^3, and the real parts of σ and v now run over intervals of length one. The exponent becomes

$$\check\sigma n+\check\rho m+\check v\ell=\sigma\frac{n}{s^2}+\rho m+v\frac{\ell}{s},$$

[5] For $\ell=1,2$ this coincidence is noted in footnote 2 of [4]; the case $\ell=0$ is the same check one coefficient further.

[6] The weight s in (7) is the function $g(s)=s$ of [6], which is fixed there by the wall crossing formula for decays into non-primitive half-BPS states. In (7) the variable $\check\sigma$ is conjugate to $Q^2/2$, $\check\rho$ to $P^2/2$ and $\check v$ to $Q\cdot P$; in the notation (2) this is $(\check\rho,\check\sigma,\check v)=(\tau',\tau,z)$, and the factor $\frac12$ in the exponent turns Q^2,P^2 into n,m.

so the integral is the Fourier coefficient of $1/\Phi_{10}(\rho,\sigma,v)$ at the invariants $(n/s^2,\ell/s,m)$. The contour (8) becomes

$$\operatorname{Im}\rho = 2n\Lambda = 2\frac{n}{s^2}\Lambda', \qquad \operatorname{Im}\sigma = 2s^2 m\Lambda = 2m\Lambda', \qquad \operatorname{Im} v = -s\ell\Lambda = -\frac{\ell}{s}\Lambda'$$

with $\Lambda' = s^2\Lambda$, which is the attractor contour of $(n/s^2,\ell/s,m)$ in Definition 2.1. Hence the term of s equals $s\,(-1)^{\ell+1}\cdot(-1)^{\ell/s+1}d^*(n/s^2,\ell/s,m)$, and $(-1)^{\ell+1}(-1)^{\ell/s+1} = (-1)^{\ell-\ell/s}$. Finally $4m(n/s^2)-(\ell/s)^2 = \Delta/s^2$. □

Remark 2.4. For $r=1$ the formula reduces to the identity, and for $r=2$ it reads $d_2^* = d^*(n,\ell,m) \pm 2\,d^*(n/4,\ell/2,m)$, with the minus sign exactly when $\ell/2$ is odd. Without the sign, (9) is equation (3.28) of [11]; the sign comes from the factor $(-1)^{Q\cdot P+1}$ in (7) against the factor $(-1)^{\ell/s+1}$ in (3).

Sen's argument applies to every single-centred black hole, so it predicts $d_r^*(Q,P) > 0$ for every $(Q,P)\in\mathcal{B}_r$ with $\Delta>0$. For $r=1$ and $m\in\{1,2\}$ this is the theorem of Bringmann and Murthy [4].

3. The reduction

Lemma 3.1. *Let $s \mid r$ and $\ell \in r\mathbb{Z}$. The sign $(-1)^{\ell-\ell/s}$ in (9) is -1 if and only if s is even and ℓ/s is odd. In that case the sign attached to $s/2$ is $+1$.*

Proof. Write $\ell = s\cdot(\ell/s)$, so $\ell-\ell/s = (s-1)(\ell/s)$. If s is odd this is even. If s is even it has the parity of ℓ/s. For the second claim, $\ell/(s/2) = 2\ell/s$ is even and ℓ is even, so $\ell-\ell/(s/2)$ is even. □

Proof of Theorem B. Let $(Q,P)\in\mathcal{B}_r$ with invariants (n,ℓ,m) and $\Delta>0$. Every triple in (9) has positive discriminant, so by hypothesis every d^* on the right is positive. If r is odd every divisor is odd, all signs are $+1$ by Lemma 3.1, and $d_r^*(Q,P)>0$.

Let r be even and let $\mathcal{N}$ be the set of divisors s of r with s even and ℓ/s odd. By Lemma 3.1 the map $s\mapsto s/2$ sends $\mathcal{N}$ injectively into the set of divisors with positive sign, and its image does not meet $\mathcal{N}$. For $s\in\mathcal{N}$ put $n' = n/s^2$, $\ell' = \ell/s$; then ℓ' is odd, $4mn'-\ell'^2 = \Delta/s^2 > 0$, and the term of $s/2$ is

$$\frac{s}{2}\,d^*\Big(\frac{4n}{s^2},\frac{2\ell}{s},m\Big) = \frac{s}{2}\,d^*(4n',2\ell',m).$$

By (1), the sum of the terms of s and $s/2$ is

$$\frac{s}{2}\,d^*(4n',2\ell',m) - s\,d^*(n',\ell',m) = \frac{s}{2}\Big(d^*(4n',2\ell',m) - 2d^*(n',\ell',m)\Big) > 0.$$

The divisors not in $\mathcal{N}\cup\frac{1}{2}\mathcal{N}$ carry positive terms. The divisor $s=1$ is never in $\mathcal{N}$, and if $1\in\frac{1}{2}\mathcal{N}$ it is paired; in either case the sum is a non-empty sum of positive quantities, so $d_r^*(Q,P)>0$. □

Remark 3.2. The pairing uses only the terms s and $s/2$, and discards all other positive terms. Figure 1 shows the pattern for $r=60$, and Figure 2 shows the sizes of the terms in three examples with $r=12$. The term $s=1$ dominates by many orders of magnitude, so (1) is far from what positivity really needs; it is simply the cleanest sufficient condition.

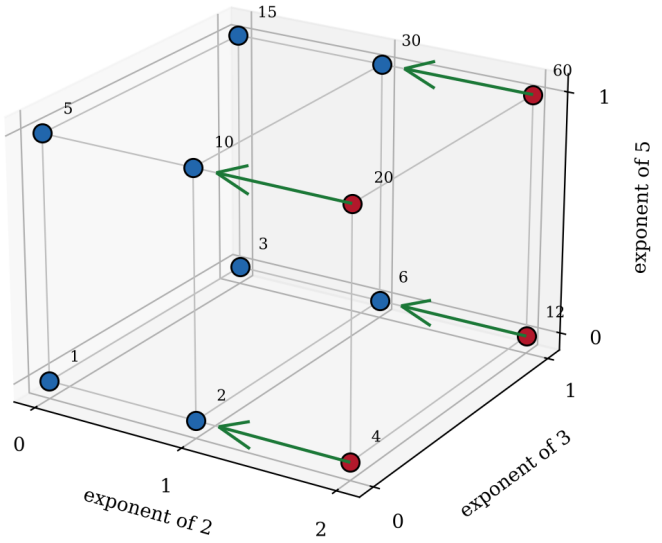


FIGURE 1. The divisors of $r = 60$ arranged by their prime exponents. If ℓ/r is odd, then ℓ/s is odd exactly when s contains the full power 4 of 2, so the negative terms of (9) sit on the face where the exponent of 2 is 2 (red). Each is paired with $s/2$ on the adjacent face (arrows), whose term is positive and has four times the discriminant.

4. Two lemmas on coloured partitions

Lemma 4.1. *For $k \geq 1$ and $N \geq 0$,*

$$p_k(N+1) \leq (k+1)\, p_k(N), \qquad p_k(N) \geq \binom{N+k-1}{k-1}, \qquad p_k(0) = 1, \quad p_k(1) = k.$$

Proof. A k-coloured partition of $N+1$ is a multiset of parts, each part a positive integer carrying one of k colours. Split the k-coloured partitions λ of $N+1$ into two classes. If λ has a part equal to 1, remove one part 1 of the smallest colour that occurs among the parts 1. The result is a partition of N, and λ is recovered from it and the colour of the removed part, so this class has at most $k\, p_k(N)$ elements. If λ has no part 1, let $a \geq 2$ be its smallest part and among the parts equal to a choose one of the smallest colour c; replace it by $a-1$ of colour c. In the result the part $a-1$ is the unique smallest part, so λ is recovered by raising it, and this class has at most $p_k(N)$ elements. Hence $p_k(N+1) \leq (k+1)p_k(N)$.

For the lower bound, the partitions using only parts equal to 1 already give $\binom{N+k-1}{k-1}$, the number of ways to distribute N ones among k colours. The values at $N = 0, 1$ are clear. □

Lemma 4.2. *Let*

$$\Psi(q) = \frac{(q^4;q^4)_\infty}{(q;q)_\infty} = \prod_{4 \nmid j} \frac{1}{1-q^j}.$$

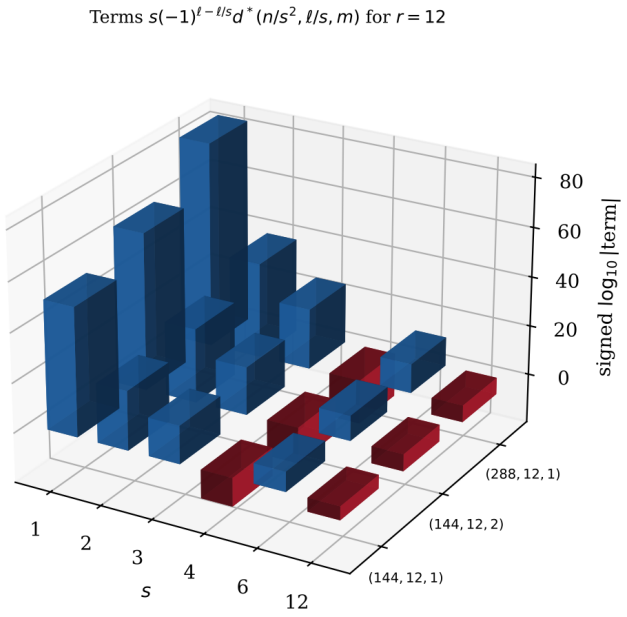


FIGURE 2. Signed size $\pm\log_{10}|t_s|$ of the terms $t_s = s(-1)^{\ell-\ell/s}d^*(n/s^2,\ell/s,m)$ of (9) for $r = 12$ and the charges $(n,\ell,m) = (144,12,1)$, $(144,12,2)$ and $(288,12,1)$. Negative terms are red; they occur at $s = 4$ and $s = 12$, where ℓ/s is odd.

Then $\Psi^k \succeq (1-q)^{-k}$, *so* $\big[\Psi^k\big]_M \geq \binom{M+k-1}{k-1}$ *for all* $M \geq 0$, *with* $\big[\Psi^k\big]_0 = 1$ *and* $\big[\Psi^k\big]_1 = k$. *Moreover*

$$P_k(q) = \Psi(q)^k\, P_k(q^4), \tag{10}$$

and for every power series F *and all integers* $N \geq 0$, $0 \leq a \leq 3$,

$$\big[F\,P_k\big]_{4N+a} = \sum_{i=-1}^{N} \big[F\Psi^k\big]_{4i+a}\, p_k(N-i), \tag{11}$$

where the term $i = -1$ *is present only when* $4i + a \geq 0$.

Proof. The product formula shows that Ψ is the generating function of partitions into parts not divisible by 4. Every $M \geq 0$ has the partition into M ones, so $\Psi \succeq (1-q)^{-1}$, and since both series have non-negative coefficients the same holds for the k-th powers. Identity (10) is $(q;q)^{-k} = \big((q^4;q^4)/(q;q)\big)^k (q^4;q^4)^{-k}$. Since $P_k(q^4) = \sum_t p_k(t) q^{4t}$,

$$\big[F\Psi^k P_k(q^4)\big]_{4N+a} = \sum_{t\geq 0} \big[F\Psi^k\big]_{4N+a-4t}\, p_k(t),$$

and the substitution $i = N - t$ gives (11); the index i is at least -1 because $4i + a \geq 0$ with $a \leq 3$. □

The point of (11) is that the coefficient of q^{4N} of a product with P_k is expressed through $p_k(N-i)$, which is comparable to the coefficient of q^N of a product with P_k. This is what allows a comparison of the two sides of (1), whose discriminants differ by a factor four.

5. Magnetic charge one

5.1. **The single-centred index.** By [4, (3.6), (4.1)],

$$\psi_1^F = \frac{1}{\Delta}\big(3E_4\,\varphi_{-2,1} - 648\,\mathcal{H}\big), \qquad \mathcal{H}(\tau,z) = \sum_{n,\ell} H(4n-\ell^2)\,q^n\zeta^\ell, \tag{12}$$

where $\Delta = \eta^{24}$ is the discriminant function[7] and $\varphi_{-2,1} = \vartheta_1(\tau,z)^2/\eta^6$ is the weak Jacobi form of weight -2 and index 1 [14]. For an integer $D>0$ with $D\equiv 0,3 \pmod 4$ let $\kappa(D)$ denote the common value of $d^*(n,\ell,1)$ over the triples with $4n-\ell^2 = D$; by Lemma 2.2 and the elliptic invariance it depends only on D.

Proposition 5.1. *For $N\geq 1$,*

$$\kappa(4N-4) = \big[6E_4TP_{30} + 648\,H_0P_{24}\big]_N, \tag{13}$$

$$\kappa(4N-1) = 3\big[E_4\theta_0P_{30}\big]_{N+1} - 648\big[H_1P_{24}\big]_N, \tag{14}$$

where $H_0 = \sum_{j\geq 0} H(4j)q^j$ and $H_1 = \sum_{j\geq 0} H(4j+3)q^j$.

The proof, a direct rewriting of (12) through theta decompositions, is in Appendix A. For example $\kappa(3) = 25353$, $\kappa(4) = 50064$, $\kappa(7) = 561576$ and $\kappa(8) = 1127472$.

For ℓ' odd and $4n'-\ell'^2 = D'$ we have $D' = 4N-1$ with $N\geq 1$, and the triple $(4n',2\ell',1)$ has discriminant $4D' = 16N-4$. So (1) for $m=1$ is the statement

$$\kappa(16N-4) > 2\,\kappa(4N-1) \qquad (N\geq 1), \tag{15}$$

and by (13) with $4N$ in place of N, $\kappa(16N-4) = \big[6E_4TP_{30} + 648H_0P_{24}\big]_{4N}$.

5.2. **The coefficients.** Put

$$c(i) = \big[E_4\theta_0\big]_i, \qquad E_4^\circ = E_4T - 225q, \qquad b(i) = \big[E_4^\circ\Psi^{30}\big]_{4i}.$$

Since $E_4\succeq 0$, $T\succeq 0$, $\theta_0\succeq 0$ and $\big[E_4T\big]_1 = 240$, we have $c(i)\geq 0$, $E_4^\circ\succeq 0$, $b(i)\geq 0$ and $b(0)=1$.

Lemma 5.2. (a) *$c(0)=1$ and $c(i)\leq 288.5\,(1+2\sqrt{i})\,i^3$ for $i\geq 1$.*

(b) *$b(i+1)\geq 961\,c(i)$ for every $i\geq 0$.*

(c) *$p_{30}(N) > 30\,c(N) + c(N+1)$ for every $N\geq 9$.*

Proof. (a) For $a\geq 1$, $\big[E_4\big]_a = 240\sigma_3(a) = 240a^3\sum_{d|a} d^{-3} \leq 240\,\zeta(3)\,a^3 < 288.5\,a^3$, since $240\zeta(3) = 288.49\ldots$. Now $c(i) = \sum_{k^2\leq i}\big[E_4\big]_{i-k^2}$ over $k\in\mathbb{Z}$. There are at most $1+2\sqrt{i}$ such k, and each term is at most $288.5\,i^3$; for $k^2=i$ the term is $1\leq 288.5\,i^3$.

(b) Because $E_4^\circ\succeq 0$ has constant term 1, $b(i+1)\geq\big[\Psi^{30}\big]_{4i+4}\geq\binom{4i+33}{29}$ by Lemma 4.2. For $i=0$ this is $40920\geq 961 = 961\,c(0)$. For $i\geq 1$ write

$$\binom{4i+33}{29} = \prod_{j=1}^{29}\frac{4i+4+j}{j} \;\geq\; \prod_{j=1}^{4}\frac{4i+4+j}{j}\cdot\prod_{j=5}^{29}\frac{8+j}{j} = \binom{4i+8}{4}\cdot 77996,$$

where we used $i\geq 1$ in the second product, and $\prod_{j=5}^{29}\frac{j+8}{j} = \frac{30\cdot 31\cdots 37}{5\cdot 6\cdots 12} = 77996$. Since $\binom{4i+8}{4}\geq (4i)^4/24$, we get $b(i+1)\geq\frac{32}{3}\cdot 77996\,i^4 > 831957\,i^4$. On the other hand, by (a) and $1+2\sqrt{i}\leq 3i$, $961\,c(i)\leq 961\cdot 288.5\cdot 3\,i^4 = 831745.5\,i^4$.[8]

[7] For the arithmetic of Δ and its coefficients see [23] and [1].

[8] The margin $831957 > 831745.5$ is narrow but sufficient; with the true values the ratio $b(i+1)/961\,c(i)$ exceeds 1500 for $0\leq i\leq 98$.

(c) By (a), and since the bound in (a) increases with i,

$$30c(N) + c(N+1) \le 31 \cdot 288.5\,(1 + 2\sqrt{N+1})(N+1)^3 \le 26830.5\,(N+1)^{7/2},$$

using $1 + 2\sqrt{x} \le 3\sqrt{x}$ for $x \ge 1$. By Lemma 4.1, $p_{30}(N) \ge \binom{N+29}{29}$. Let $f(N) = \binom{N+29}{29}(N+1)^{-7/2}$. Then

$$\frac{f(N+1)}{f(N)} = \Big(1 + \frac{29}{N+1}\Big)\Big(1 + \frac{1}{N+1}\Big)^{-7/2} > 1$$

for $N \ge 1$, because $(1+x)^{7/2} \le e^{7x/2} \le 1 + \frac{7}{2}xe^{7x/2} \le 1 + \frac{7}{2}e^{7/4}x < 1 + 20.2x$ for $0 < x \le \frac{1}{2}$. Since $f(9) = \binom{38}{29}/10^{7/2} = 163011640/10^{7/2} > 51548$, we get $p_{30}(N) \ge f(N)(N+1)^{7/2} > 26830.5\,(N+1)^{7/2}$ for $N \ge 9$. □

Theorem 5.3. *Inequality* (15) *holds for every* $N \ge 1$. *Hence* (1) *holds for* $m = 1$.

Proof. The right side. Since $H(4j+3) > 0$, (14) gives

$$2\kappa(4N-1) \le 6\big[E_4\theta_0 P_{30}\big]_{N+1} = 6\sum_{i=0}^{N+1} c(i)\,p_{30}(N+1-i).$$

The left side. The series $648H_0$ has constant term $648H(0) = -54$ and all other coefficients positive, so $648\,H_0P_{24} \succeq -54P_{24}$ and

$$\kappa(16N-4) \ge 6\big[E_4TP_{30}\big]_{4N} - 54\,p_{24}(4N).$$

By Lemma 4.1, $p_{24}(4N) \le 25\,p_{24}(4N-1)$, and $p_{24}(4N-1) \le p_{30}(4N-1)$ because $P_{30} = P_{24}P_6$ with $P_6 \succeq 0$ of constant term 1. Hence $54\,p_{24}(4N) \le 1350\,p_{30}(4N-1) = 6 \cdot 225\big[qP_{30}\big]_{4N}$, and

$$\kappa(16N-4) \ge 6\big[E_4^\circ P_{30}\big]_{4N} = 6\sum_{i=0}^{N} b(i)\,p_{30}(N-i)$$

by (11) with $a = 0$ (the term $i = -1$ is absent).

Comparison. For $0 \le i \le N-1$, Lemma 4.1 applied twice and Lemma 5.2(b) give

$$c(i)\,p_{30}(N+1-i) \le 31^2\,c(i)\,p_{30}(N-1-i) \le b(i+1)\,p_{30}(N-1-i).$$

The right side is the term $i+1$ of the sum for $\kappa(16N-4)/6$. Summing over $0 \le i \le N-1$, the terms $i = N, N+1$ of the sum for $2\kappa(4N-1)/6$ and the term $i = 0$ of the sum for $\kappa(16N-4)/6$ are left over, and with $b(0) = 1$, $p_{30}(1) = 30$, $p_{30}(0) = 1$ we get

$$\frac{\kappa(16N-4) - 2\kappa(4N-1)}{6} \;\ge\; p_{30}(N) - 30\,c(N) - c(N+1).$$

For $N \ge 9$ the right side is positive by Lemma 5.2(c). For $1 \le N \le 8$, (15) is checked directly from the exact values in Table 2. □

6. Magnetic charge two

6.1. **The single-centred index.** By [4, (3.7)],

$$\psi_2^F = \frac{1}{3\Delta}\Big(22E_4\varphi_{-2,1}\varphi_{0,1} - 10E_6\varphi_{-2,1}^2 - 9600\,\mathcal{H}|V_2\Big), \tag{16}$$

where $\varphi_{0,1}$ is the weak Jacobi form of weight 0 and index 1 and V_2 is the Hecke-type operator of [14]. Let

$$\begin{aligned} g &= 5E_6 - 11E_2E_4, \\ \phi_0 &= (66T + 264\,DT)E_4 + gT, \qquad \phi_1 = 264\,(D\theta_0)E_4 + g\theta_0, \\ R &= 8q\,T(q^2)\,T\,\phi_0 + \theta_0(q^2)\,\theta_0\,\phi_1, \qquad S = 2\phi_0\theta_0 + 2T\phi_1, \end{aligned}$$

and write $R = \sum R(a)q^a$, $S = \sum S(a)q^a$. Then

$$\begin{aligned} R &= -6 - 3912q + 152940q^2 + 2070576q^3 + 12282600q^4 + 54725808q^5 + \cdots, \\ S &= 108 + 13368q + 392844q^2 + \cdots. \end{aligned}$$

Bringmann and Murthy proved that $R(a) > 0$ for all $a \geq 2$ and that $S \succeq 0$ [4, Section 6].[9]

For $D > 0$ with $D \equiv 7 \pmod 8$ let $L_1(D)$ be the common value of $d^*(n, \ell, 2)$ over ℓ odd and $8n - \ell^2 = D$, and for $D \equiv 4 \pmod 8$ let $L_2(D)$ be the common value over $\ell \equiv 2 \pmod 4$. By Lemma 2.2 these are well defined.

Proposition 6.1. *Let $D' \equiv 7 \pmod 8$, $D' > 0$, and put $K = (D'+3)/2$ and $M = (D'+9)/8$. Then*

$$L_2(4D') = \tfrac{2}{3}\Big(\big[RP_{36}\big]_K + 4800\big[C_2P_{24}\big]_{K-1}\Big), \tag{17}$$

$$L_1(D') = \tfrac{2}{3}\Big(\big[VSP_{36}\big]_M - 4800\big[H_1'P_{24}\big]_M\Big), \tag{18}$$

where $C_2 = \sum_{k\geq 0}\big(H(8k+4) + 2H(2k+1)\,[k \text{ odd}]\big)q^k$ and $H_1' = \sum_{j\geq 1} H(8j-1)q^j$.

The proof is in Appendix A. For example $L_1(7) = 561576$ and $L_2(28) = 2073849984$.

For ℓ' odd and $8n' - \ell'^2 = D'$ we have $D' \equiv 7 \pmod 8$, and $2\ell' \equiv 2 \pmod 4$. Thus (1) for $m = 2$ is

$$L_2(4D') > 2L_1(D') \qquad (D' = 8j+7,\ j \geq 0). \tag{19}$$

In terms of j we have $K = 4j+5$ and $M = j+2$.

6.2. **The coefficients.** Put

$$R_{\geq 4} = \sum_{a\geq 4} R(a)q^a, \qquad \rho(i) = \big[R_{\geq 4}\Psi^{36}\big]_{4i+5}, \qquad \sigma(i) = \big[VS\big]_i.$$

Then $R_{\geq 4} \succeq 0$, so $\rho(i) \geq 0$, and $\sigma(i) \geq 0$. Since $\big[\Psi^{36}\big]_1 = 36$,

$$\rho(0) = 36R(4) + R(5) = 496899408, \qquad \sigma(0) = S(0) = 108.$$

Lemma 6.2. (a) *$\sigma(i) \leq 5.5164 \cdot 10^6\, i^{13/2}$ for $i \geq 1$.*

(b) *$\rho(i) \geq 2738\,\sigma(i)$ for every $i \geq 0$.*

(c) *$\big(\rho(0) - 2738 \cdot 108\big)\,p_{36}(j) > 2\big(36\,\sigma(j+1) + \sigma(j+2)\big)$ for every $j \geq 5$.*

Proof. (a) is Lemma B.3 in Appendix B.

(b) For $i = 0$: $\rho(0) = 496899408 > 2738 \cdot 108 = 295704$. For $i \geq 1$, keeping only the term $R(4)q^4$,

$$\rho(i) \geq R(4)\big[\Psi^{36}\big]_{4i+1} \geq R(4)\binom{4i+36}{35}$$

[9] In the notation of [4], $R = -\eta^{12}\xi_2$ and $S = q^{-1/4}(f_0\theta_0 + f_1\theta_1)$; see Appendix A. Their argument estimates g from below by $2529a^5 - 7920a^4$ and controls the theta products by their Lemma 6.1.

by Lemma 4.2. Write

$$\binom{4i+36}{35} = \prod_{j=1}^{35} \frac{4i+1+j}{j} \;\geq\; \frac{(4i)^7}{7!} \prod_{j=8}^{35} \frac{5+j}{j} = \frac{(4i)^7}{7!} \cdot \frac{36\cdot 37\cdot 38\cdot 39\cdot 40}{8\cdot 9\cdot 10\cdot 11\cdot 12} = \frac{16384\, i^7}{5040} \cdot \frac{9139}{11},$$

where in the second product we used $i \geq 1$. With $R(4) = 12282600$ this gives $\rho(i) > 3.31\cdot 10^{10}\, i^7$. By (a), $2738\,\sigma(i) \leq 1.5104 \cdot 10^{10}\, i^{13/2}$.

(c) By (a), $2(36\sigma(j+1)+\sigma(j+2)) \leq 2\cdot 37\cdot 5.5164\cdot 10^6\,(j+2)^{13/2} < 4.0822\cdot 10^8\,(j+2)^{13/2}$, while $\rho(0) - 295704 = 496603704$ and $p_{36}(j) \geq \binom{j+35}{35}$. Let $h(j) = \binom{j+35}{35}(j+2)^{-13/2}$. For $j \geq 5$,

$$\frac{h(j+1)}{h(j)} = \Big(1+\frac{35}{j+1}\Big)\Big(1+\frac{1}{j+2}\Big)^{-13/2} > 1,$$

since with $x = 1/(j+2) \leq \frac{1}{7}$ we have $(1+x)^{13/2} \leq 1 + \frac{13}{2}xe^{13x/2} \leq 1 + \frac{13}{2}e^{13/14}x < 1 + 16.5x$, and $16.5/(j+2) < 35/(j+1)$. Finally $h(5) = 658008/7^{13/2} > 2.11$, and $496603704 \cdot 2.11 > 1.04\cdot 10^9 > 4.0822\cdot 10^8$, so $496603704\, p_{36}(j) \geq 496603704\, h(5)(j+2)^{13/2} > 4.0822\cdot 10^8 (j+2)^{13/2}$ for $j \geq 5$. □

Theorem 6.3. *Inequality* (19) *holds for every* $j \geq 0$. *Hence* (1) *holds for* $m = 2$.

Proof. Dropping the class numbers. The coefficients of C_2 and H_1' are sums of Hurwitz class numbers $H(N)$ with $N \geq 1$, hence non-negative. By (17) and (18) it suffices to show

$$\big[RP_{36}\big]_{4j+5} > 2\big[VSP_{36}\big]_{j+2}. \tag{20}$$

The first four terms of R. Let $K = 4j+5 \geq 5$. By Lemma 4.1, $p_{36}(K) \leq 37 p_{36}(K-1)$ and $p_{36}(K-1) \leq 37 p_{36}(K-2)$, so

$$6p_{36}(K) + 3912\, p_{36}(K-1) \leq (6\cdot 37 + 3912)\, p_{36}(K-1) = 4134\, p_{36}(K-1) \leq 152958\, p_{36}(K-2).$$

Since $R(2) = 152940$, the deficit is $18\, p_{36}(K-2) \leq 666\, p_{36}(K-3) \leq R(3)\, p_{36}(K-3)$. Hence

$$\sum_{a=0}^{3} R(a)\, p_{36}(K-a) \geq 0, \qquad \text{so} \qquad \big[RP_{36}\big]_K \geq \big[R_{\geq 4}P_{36}\big]_K.$$

Dilation. By (11) with $a = 1$ and $N = j+1$,

$$\big[R_{\geq 4}P_{36}\big]_{4j+5} = \sum_{i=-1}^{j+1} \big[R_{\geq 4}\Psi^{36}\big]_{4i+5}\, p_{36}(j+1-i) \;\geq\; \sum_{i=0}^{j} \rho(i)\, p_{36}(j-i),$$

since after shifting i by one the sum is $\sum_{i=0}^{j+1} \rho(i)\, p_{36}(j+1-i)$ (the term with $\big[R_{\geq 4}\Psi^{36}\big]_1 = 0$ drops out), and $p_{36}(j+1-i) \geq p_{36}(j-i)$. On the other side,

$$2\big[VSP_{36}\big]_{j+2} = 2\sum_{i=0}^{j+2} \sigma(i)\, p_{36}(j+2-i).$$

Comparison. For $0 \leq i \leq j$, Lemma 4.1 applied twice gives $p_{36}(j+2-i) \leq 37^2 p_{36}(j-i) = 1369\, p_{36}(j-i)$, so by Lemma 6.2(b)

$$2\sigma(i)\, p_{36}(j+2-i) \leq 2738\,\sigma(i)\, p_{36}(j-i) \leq \rho(i)\, p_{36}(j-i).$$

For $i = 0$ we keep the exact value $2738\,\sigma(0) = 295704$. Summing,

$$\big[R_{\geq 4}P_{36}\big]_{4j+5} - 2\big[VSP_{36}\big]_{j+2} \;\geq\; \big(\rho(0) - 295704\big)p_{36}(j) - 2\big(36\,\sigma(j+1) + \sigma(j+2)\big),$$

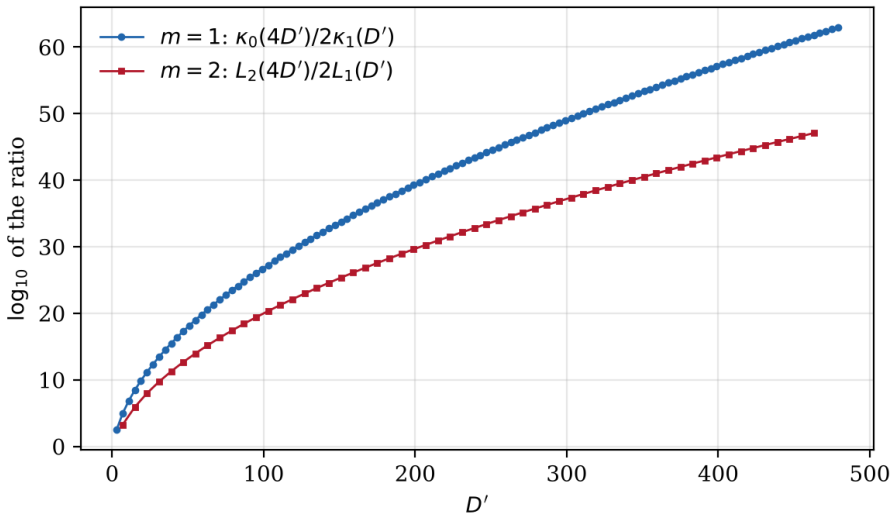


FIGURE 3. The ratio of the two sides of the doubling inequality for $m = 1$ (as a function of $D' = 4N - 1 \le 479$) and for $m = 2$ (as a function of $D' = 8j + 7 \le 463$), on a logarithmic scale. The smallest ratios are 325 at $D' = 3$ and 1846 at $D' = 7$; the ratio grows roughly like $\exp(\pi\sqrt{D'})$.

using $p_{36}(1) = 36$ and $p_{36}(0) = 1$. For $j \ge 5$ the right side is positive by Lemma 6.2(c). For $0 \le j \le 4$, that is $D' \in \{7, 15, 23, 31, 39\}$, inequality (19) is checked from the exact values in Table 3. □

7. Proof of Theorem A, and general magnetic charge

Proof of Theorems A and C. Theorem C is Theorems 5.3 and 6.3. For $m \in \{1, 2\}$ Bringmann and Murthy proved that $(-1)^{\ell+1} c_m^F(n, \ell) > 0$ whenever $4mn - \ell^2 > 0$ [4, (3.8)]. By Lemma 2.2 this says $d^*(n', \ell', m) > 0$ for all (n', ℓ') with positive discriminant, which is the hypothesis of Theorem B. Theorem C supplies (1), and Theorem B gives $d_r^*(Q, P) > 0$. □

Example 7.1. Take $r = 2$, $m = 1$ and $(n, \ell) = (4, 2)$, so $\Delta = 12$. Then $\ell/2 = 1$ is odd and

$$d_2^* = d^*(4, 2, 1) - 2\, d^*(1, 1, 1) = \kappa(12) - 2\kappa(3).$$

By (13) and (14), $\kappa(3) = 25353$ and $\kappa(12) = 16491600$, so $d_2^* = 16440894 > 0$. The ratio $\kappa(12)/2\kappa(3) \approx 325.2$ is the first row of Table 2, and it is the smallest doubling ratio we have met for any m (Proposition 7.2).

7.1. **General magnetic charge.** For $m \ge 3$ the positivity of the primitive single-centred indices is not part of this paper, and Theorem B reduces the case of torsion r to it, unconditionally for odd r and through (1) for even r. We computed the finite parts ψ_m^F exactly from the product formula for Φ_{10}, by expanding the Fourier–Jacobi coefficients ψ_m and subtracting the Appell–Lerch polar parts, and evaluated d^* at reduced triples through (5).

Proposition 7.2 (computation)**.** *The doubling inequality* (1) *holds for all* 2410 *triples* (n', ℓ', m) *with* ℓ' *odd,* $|\ell'| \le m$*,* $m \le 199$ *and* $4\Delta' \le 240$*; the smallest ratio of the two sides is* 325.2*, attained at* $(1, \pm 1, 1)$*. The single-centred index* (9) *is positive for all* 6065 *triples with* $2 \le r \le 15$*,* $|\ell| \le m$ *and* $\Delta \le 240$*.*

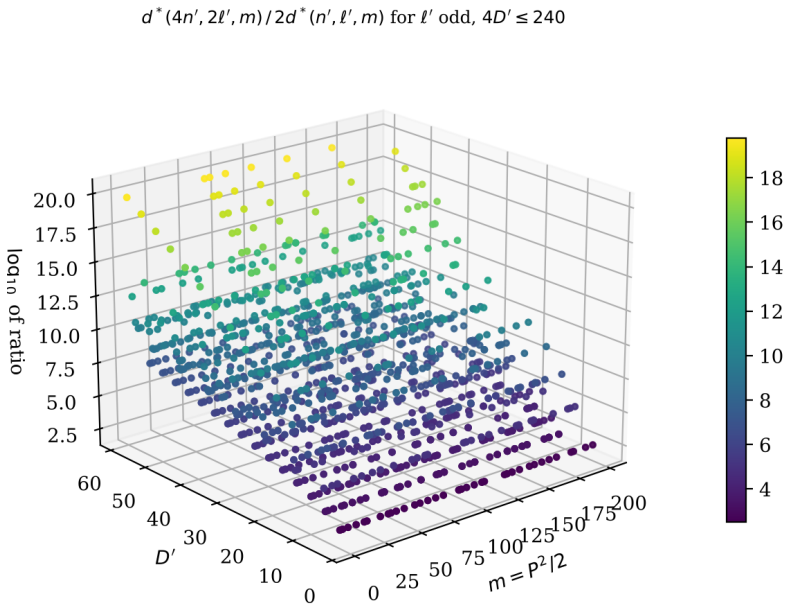


FIGURE 4. The ratio $d^*(4n', 2\ell', m)/2d^*(n', \ell', m)$ on a logarithmic scale for all 2410 triples of Proposition 7.2, plotted against m and $D' = 4mn' - \ell'^2$. The ratio depends essentially on D' alone, as expected from the growth $\exp(\pi\sqrt{\Delta})$ of the indices.

Figure 4 suggests that for fixed D' the ratio in (1) hardly depends on m. This is what one expects from the Rademacher expansion of the single-centred index, whose leading term depends only on the discriminant [17], and it suggests that (1) can be proved for all m by the same circle-method estimates that control the primitive indices. We do not pursue this here.

APPENDIX A. THE q-SERIES FOR $m = 1$ AND $m = 2$

We use the theta functions $\vartheta_{1,\mu}(\tau, 0)$ and $\vartheta_{2,\mu}(\tau, 0)$ of [4, 14]. For index one,

$$\theta_\mu(\tau) := \vartheta_{1,\mu}(\tau, 0) = \sum_{r \equiv \mu\,(2)} q^{r^2/4}, \qquad \theta_0 = \sum_{k \in \mathbb{Z}} q^{k^2}, \qquad \theta_1 = \sum_{r \text{ odd}} q^{r^2/4} = 2q^{1/4}T.$$

Moreover $\theta_1(\tau/2) = \sum_{r \text{ odd}} q^{r^2/8} = 2q^{1/8}V$ and $\theta_1(2\tau) = 2q^{1/2}T(q^2)$. Writing $\theta' = D\theta$, we have $\theta_1' = \sum_{r \text{ odd}} \frac{r^2}{4} q^{r^2/4}$, and since $\frac{(2k+1)^2}{4} = k(k+1) + \frac{1}{4}$,

$$\theta_1' = 2q^{1/4}\big(DT + \tfrac{1}{4}T\big). \tag{21}$$

A.1. **Proof of Proposition 5.1.** Bringmann and Murthy write [4, Section 4]

$$\psi_1^F = k_1\vartheta_{1,1} - k_0\vartheta_{1,0}, \qquad k_1 = \frac{3}{\Delta}\Big(\frac{E_4\theta_0}{\eta^6} - 216\,\mathcal{H}_1\Big), \qquad k_0 = \frac{3}{\Delta}\Big(\frac{E_4\theta_1}{\eta^6} + 216\,\mathcal{H}_0\Big),$$

with $\mathcal{H}_\mu = \sum_{j \ge 0} H(4j+3\mu)q^{j+3\mu/4}$, so that $\mathcal{H}_0 = H_0$ and $\mathcal{H}_1 = q^{3/4}H_1$. The coefficient $c_1^F(n, \ell)$ is the coefficient of $q^{(4n-\ell^2)/4}$ in k_1 for ℓ odd and in $-k_0$ for ℓ even, so $\kappa(D) = (-1)^{\ell+1}c_1^F(n, \ell)$ is the coefficient of $q^{D/4}$ in k_1 when $D \equiv 3 \pmod 4$ and in k_0 when $D \equiv 0 \pmod 4$.

Using $\eta^{30} = q^{5/4}(q;q)_\infty^{30}$ and $\Delta = q\,(q;q)_\infty^{24}$,

$$k_0 = \frac{3E_4 \cdot 2q^{1/4}T}{q^{5/4}(q;q)_\infty^{30}} + \frac{648H_0}{q\,(q;q)_\infty^{24}} = q^{-1}\big(6E_4TP_{30} + 648H_0P_{24}\big).$$

The coefficient of $q^{D/4}$ with $D = 4N-4$ is the coefficient of q^N of the bracket, which is (13). Similarly

$$k_1 = 3q^{-5/4}E_4\theta_0P_{30} - 648\,q^{-1/4}H_1P_{24},$$

and the coefficient of $q^{D/4}$ with $D = 4N-1$, that is of $q^{N-1/4}$, is $3\big[E_4\theta_0P_{30}\big]_{N+1} - 648\big[H_1P_{24}\big]_N$, which is (14).

A.2. **Proof of Proposition 6.1.** Bringmann and Murthy define ξ_μ by $11E_4\varphi_{-2,1}\varphi_{0,1} - 5E_6\varphi_{-2,1}^2 = \sum_{\mu=0}^3 \xi_\mu\vartheta_{2,\mu}$ and F_μ by $\mathcal{H}|V_2 = \sum_\mu F_\mu\vartheta_{2,\mu}$ [4, (5.11), (6.1)]. By (16),

$$\psi_2^F = \sum_\mu \frac{2(\xi_\mu - 4800F_\mu)}{3\Delta}\vartheta_{2,\mu},$$

so for $\ell \equiv \mu \pmod 4$ the index $d^*(n,\ell,2) = (-1)^{\ell+1}c_2^F(n,\ell)$ is the coefficient of $q^{(8n-\ell^2)/8}$ in $\frac{2}{3}(-1)^{\mu+1}(\xi_\mu - 4800F_\mu)/\Delta$. Their formulas [4, (6.4), (6.5)] read

$$\eta^{12}\xi_1 = \tfrac{1}{2}\theta_1(\tau/2)\,(f_0\theta_0 + f_1\theta_1), \qquad -\eta^{12}\xi_2 = \theta_1(2\tau)\theta_1f_0 + \theta_0(2\tau)\theta_0f_1,$$

with $f_0 = 264\theta_1'E_4 + g\theta_1$ and $f_1 = 264\theta_0'E_4 + g\theta_0$ [4, (5.2), (5.3)]. By (21), $f_0 = 2q^{1/4}\big((66T + 264DT)E_4 + gT\big) = 2q^{1/4}\phi_0$, and $f_1 = \phi_1$. Hence

$$f_0\theta_0 + f_1\theta_1 = 2q^{1/4}(\phi_0\theta_0 + T\phi_1) = q^{1/4}S,$$
$$-\eta^{12}\xi_2 = 2q^{1/2}T(q^2)\cdot 2q^{1/4}T\cdot 2q^{1/4}\phi_0 + \theta_0(q^2)\theta_0\phi_1 = R,$$

and $\eta^{12}\xi_1 = \frac{1}{2}\cdot 2q^{1/8}V\cdot q^{1/4}S = q^{3/8}VS$.

The case $\mu = 2$. Since $\eta^{36} = q^{3/2}(q;q)_\infty^{36}$, we get $-\xi_2/\Delta = q^{-3/2}RP_{36}$. The class number part is $F_2 = \sum_{n\ge1} c(n,2)q^{n-1/2}$ with $c(n,2) = H(8n-4) + 2H(2n-1)[n \text{ even}]$ [4, (5.11)], so $F_2/\Delta = q^{-1/2}C_2P_{24}$ with $C_2 = \sum_{k\ge0} c(k+1,2)q^k$. Since $(-1)^{\mu+1} = -1$,

$$L_2(D) = \tfrac{2}{3}\Big(\big[RP_{36}\big]_{D/8+3/2} + 4800\big[C_2P_{24}\big]_{D/8+1/2}\Big),$$

and $D = 4D'$ gives $D/8 + 3/2 = K$ and $D/8 + 1/2 = K-1$, which is (17).

The case $\mu = 1$. We have $\xi_1/\Delta = q^{3/8-3/2}VSP_{36} = q^{-9/8}VSP_{36}$, and $F_1 = \sum_{j\ge1} H(8j-1)q^{j-1/8}$, so $F_1/\Delta = q^{-9/8}H_1'P_{24}$. With $(-1)^{\mu+1} = 1$, the coefficient of $q^{D'/8}$ is the coefficient of $q^{(D'+9)/8} = q^M$, which is (18).

Appendix B. Coefficient bounds

Throughout, $\zeta(3) = 1.2020569\ldots$ and $\zeta(5) = 1.0369277\ldots$, and we write $e_4(a) = \big[E_4\big]_a$.

Lemma B.1. *$g(0) = -6$ and $g(a) = 3024\,\sigma_5(a) - 7920\,a\,\sigma_3(a)$ for $a \ge 1$. In particular $|g(a)| \le 9520.3\,a^5$ for $a \ge 1$.*

Proof. Ramanujan's identity $DE_4 = \frac{1}{3}(E_2E_4 - E_6)$ [23, Section 5.2] gives $E_2E_4 = E_6 + 3DE_4$, hence $g = 5E_6 - 11E_6 - 33DE_4 = -6E_6 - 33DE_4$. The coefficient of q^a is $3024\sigma_5(a) - 33\cdot 240\,a\sigma_3(a)$. Both terms are non-negative, so $|g(a)|$ is at most the larger of them, and $3024\sigma_5(a) \le 3024\zeta(5)a^5 < 3136\,a^5$ while $7920a\sigma_3(a) \le 7920\zeta(3)a^4 < 9520.3\,a^4$. □

Lemma B.2. *For $n \geq 1$,*

$$|\phi_0(n)| \leq 104726\,(1+\sqrt{n})\,n^5, \qquad |\phi_1(n)| \leq 85685\,(1+2\sqrt{n})\,n^5,$$

and $\phi_0(0) = 60$, $\phi_1(0) = -6$.

Proof. The coefficient of q^n in ϕ_0 is $\sum_t \big((66+264t)e_4(n-t)+g(n-t)\big)$ over $t = k(k+1) \leq n$ with $k \geq 0$; there are at most $1+\sqrt{n}$ such t. If $a = n-t \geq 1$, then $(66+264t)e_4(a) \leq 330n \cdot 288.5n^3 = 95205n^4$ and $|g(a)| \leq 9520.3n^5$, so the term is at most $104725.3\,n^5$. If $t = n$, the term is $66+264n-6 \leq 330n$. This gives the bound for ϕ_0, and $\phi_0(0) = 66-6$.

Similarly the coefficient of q^n in ϕ_1 is $\sum_{k\in\mathbb{Z},\,k^2\leq n} \big(264k^2 e_4(n-k^2)+g(n-k^2)\big)$, with at most $1+2\sqrt{n}$ terms, each bounded by $264n \cdot 288.5n^3 + 9520.3n^5 \leq 85685n^5$ when $n-k^2 \geq 1$ and by $264n$ when $k^2 = n$. Finally $\phi_1(0) = g(0) = -6$. □

Lemma B.3. *For $n \geq 1$, $|S(n)| \leq 380822\,(1+\sqrt{n})(1+2\sqrt{n})\,n^5$, and for $i \geq 1$, $\sigma(i) \leq 5.5164 \cdot 10^6\, i^{13/2}$.*

Proof. We have $S(n) = 2\sum_{k\in\mathbb{Z},\,k^2\leq n} \phi_0(n-k^2) + 2\sum_{k\geq 0,\,k(k+1)\leq n} \phi_1(n-k(k+1))$. The bounds of Lemma B.2 increase with n and also dominate $|\phi_0(0)| = 60$ and $|\phi_1(0)| = 6$, so

$$\begin{aligned} |S(n)| &\leq 2(1+2\sqrt{n}) \cdot 104726(1+\sqrt{n})n^5 + 2(1+\sqrt{n}) \cdot 85685(1+2\sqrt{n})n^5 \\ &= 380822(1+\sqrt{n})(1+2\sqrt{n})n^5. \end{aligned}$$

Next $\sigma(i) = \sum_{k\geq 0,\,k(k+1)/2\leq i} S(i-k(k+1)/2)$ has at most $1+\sqrt{2i}$ terms, each at most $380822(1+\sqrt{i})(1+2\sqrt{i})i^5$ in absolute value (the term $S(0) = 108$ included). For $i \geq 1$, $(1+\sqrt{2i})(1+\sqrt{i})(1+2\sqrt{i}) \leq (1+\sqrt{2}) \cdot 2 \cdot 3\,i^{3/2} < 14.4854\,i^{3/2}$, and $380822 \cdot 14.4854 < 5.5164 \cdot 10^6$. □

Remark B.4. The constants are far from optimal. The largest value of $|g(a)|/a^5$ is 4896, at $a = 1$, and the largest value of $|S(n)|/((1+2\sqrt{n})^2 n^5)$ for $1 \leq n \leq 400$ is below 1500. Sharper constants would shorten the tables of Appendix C but are not needed.

Appendix C. Exact values

All values below are exact integers, computed from Propositions 5.1 and 6.1 with $H(N)$ obtained by counting reduced binary quadratic forms. Every value that also appears in the coefficient tables of [4, Appendix A] agrees with it.

ℓ	$(-1)^{\ell+1}c_2^F(1,\ell) = (-1)^{\ell+1}c_1^F(2,\ell)$
0	1127472
±1	561576
±2	50064

Table 1. The coefficients used in the proof of Lemma 2.2.

Declarations

Data availability. The programs that compute the q-series of Propositions 5.1 and 6.1, check every constant of Lemmas 5.2, 6.2 and Appendix B, produce Tables 1–3 and Proposition 7.2, and draw the figures are archived at [3].

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N	$D' = 4N - 1$	$\kappa(16N-4)$	$\kappa(4N-1)$
1	3	16491600	25353
2	7	101777516400	561576
3	11	120407289152016	8533821
4	15	56968670205288432	100390104
5	19	14565081067453280016	977183520
6	23	2367012428714552500416	8203464720
7	27	270323327028195388932480	61077837780
8	31	232092965138092776747995 04	411421124040
9	35	157184091664000868374166 0160	2544746970243

TABLE 2. Inequality (15) for $1 \leq N \leq 9$.

j	$D' = 8j + 7$	$L_2(4D')$	$L_1(D')$
0	7	2073849984	561576
1	15	30312295881600	18458000
2	23	73625812211910912	392427528
3	31	65440347323367281280	6216536784
4	39	29554733204711345826304	79330416536
5	47	81025154914182624278941 44	855667882536

TABLE 3. Inequality (19) for $0 \leq j \leq 5$.

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